

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)		[Freya is incorrect,] there is an anomaly (1) and should not be included (1) Award 1 mark for $\frac{7.6 + 8.0}{2} = 7.8$ [s]			2	2	1	2
		(ii)		Data is not repeatable because repeats are quite far apart (1) OR It is repeatable because the repeats are close to each other (1) OR It is repeatable apart from 20 cm (1)			1	1		1
	(b)			Timing [moving object] or reaction time or difficult to {start / stop} the stopwatch on time (1) Accept human error only if qualified Use a slow-motion camera or increase the distance travelled (1) accept record it / video it / light gates. Don't accept use lasers or use your phone Accept reference to keeping the volume of oil constant OWTTE		1	1	2		2
	(c)	(i)		Between 6.8 [s] and 7.2 [s]		1		1	1	1
		(ii)		$\frac{53 - 30}{12 - 9}$ (1) = 7.67 or 7.7 [cm/s] (1)		2		2	2	2
		(iii)		Decreases (1) To zero (1) Alternative: Starts high (1) Then it stops accelerating (1)		2		2		2
				Question 10 total	0	6	4	10	4	10